LAB - 8

AIM – DRAW A STATE DIAGRAM FOR THE PROJECT

TEAM MEMBERS -

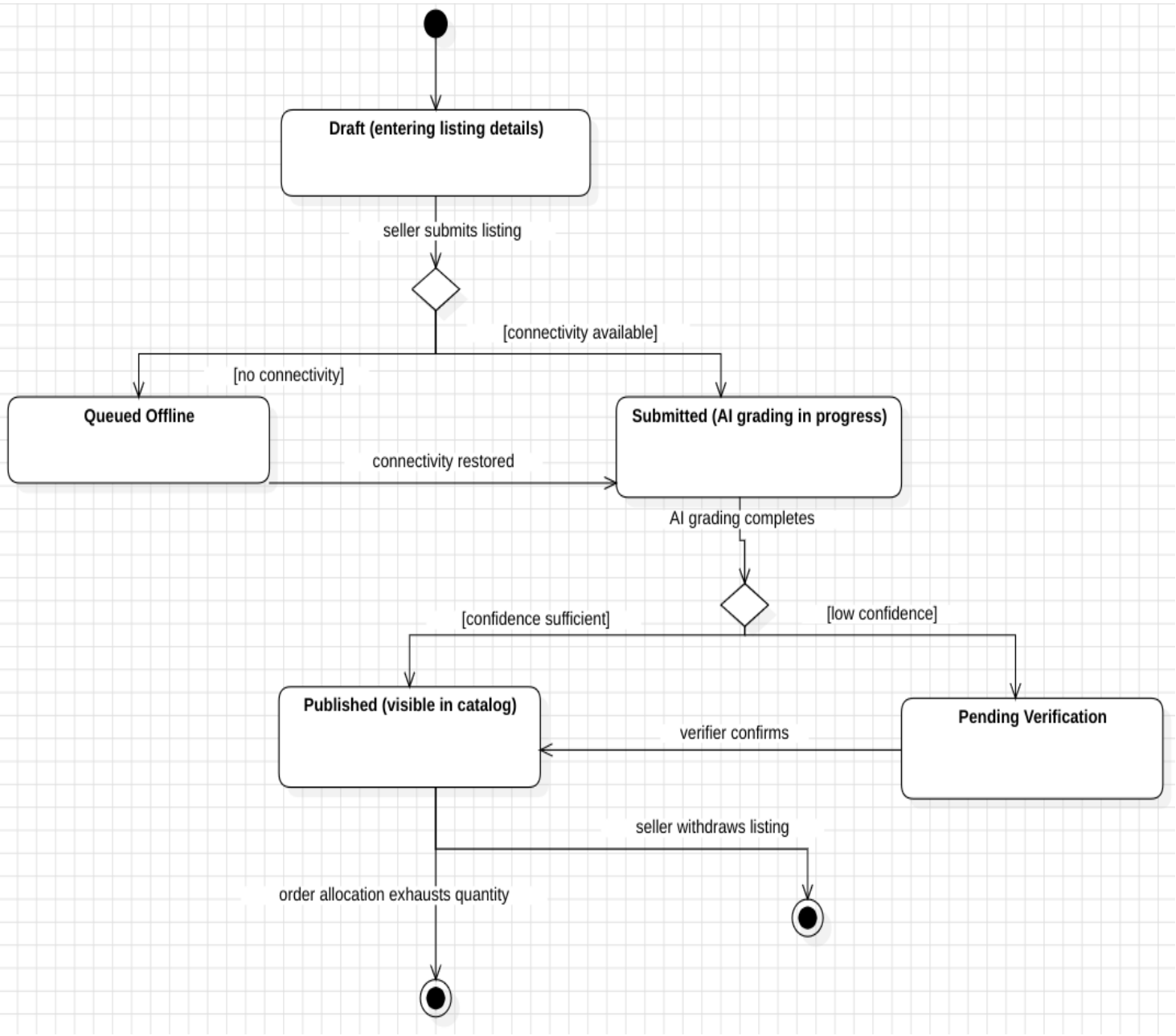
ARYAN RAJ (24BDS1094)

YASH RAJ (24BDS1209)

- **We have separated the state transition diagram for our project in 2 parts for**
  i. **Listing a product on the platform**
  ii. **Ordering System of the platform**

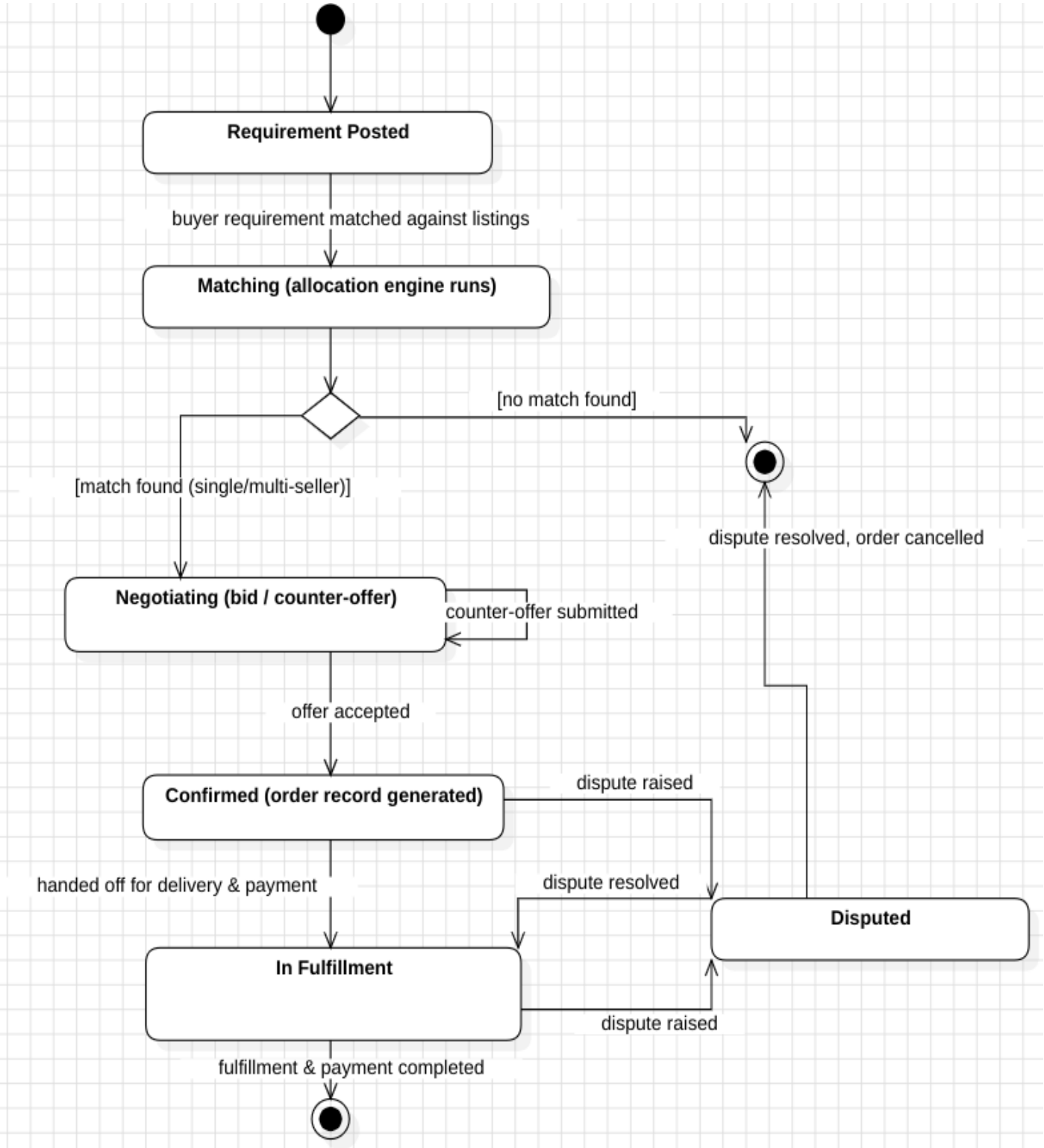
1. Listing Lifecycle (State Diagram)

1. This is a diagram showing the various states through which a listing passes from its creation to the point of conclusion.
2. The process begins with the seller being in the Draft state, providing listing details such as commodity, quantity, and location.
3. Once submitted, the system checks connectivity:
   - If there is no connectivity, the listing moves to the Queued Offline state where it waits.
   - Upon restoration of connectivity, it proceeds to Submitted state automatically.
   - If there is connectivity, it moves directly to Submitted.
4. At the Submitted stage, automatic AI-assisted grading takes place.
5. After grading, the system checks the confidence score:
   - If the confidence is adequate, the listing moves directly to Published.
   - If the confidence level is inadequate, it moves to the Pending Verification stage where it is manually graded by a human verifier prior to moving to the Published stage.
6. At the Published stage, the listing becomes visible to the buyer in the catalog.
7. From here, the listing can end in one of two final states:
   - Sold Out – when the entire quantity of the item is sold through orders.
   - Delisted – when the seller delists the listing.


Draft (entering listing details)
seller submits listing
[connectivity available]
[no connectivity]
Queued Offline
Submitted (AI grading in progress)
connectivity restored
AI grading completes
[confidence sufficient]
[low confidence]
Published (visible in catalog)
Pending Verification
verifier confirms
seller withdraws listing
order allocation exhausts quantity

2. Order Lifecycle (State Diagram)

1. The below figure represents the various stages that the order passes through, starting from requirement till its final destination.
2. The first stage is "Requirement Posted" where the requirements of the buyer are mentioned.
3. After the above step, the order enters into the "Matching" stage where the allocation engine in the system attempts to match the requirements of the buyer with available listing(s) (from multiple sellers).
4. After matching, there are two possibilities:
    - If a match is found, the order moves to Negotiating.
    - If no match is found, the order moves to Cancelled (a final state).
5. Negotiation status involves the interaction between the buyer and seller (offers and counter-offers) – a self-loop on the order, i.e. it remains in the current status while offers are exchanged.
6. Upon acceptance of the offer, the order acquires the status of Confirmed when the order record is created.
7. Upon confirmation of the order, the order receives status In Fulfillment which means that it is transferred for fulfillment and payment.
8. n any state (either Confirmed or In Fulfillment), there is an option to start a dispute, which means that the order becomes Disputed.
9. From Disputed, the order can either:
    - Return to In Fulfillment in case of successful dispute resolution in favor of keeping the order.
    - Move to Cancelled if the dispute results in cancellation.
10. After successful fulfillment and payment, the order becomes Closed (Final state).


Requirement Posted
buyer requirement matched against listings
Matching (allocation engine runs)
[no match found]
[match found (single/multi-seller)]
dispute resolved, order cancelled
Negotiating (bid / counter-offer)
counter-offer submitted
offer accepted
Confirmed (order record generated)
dispute raised
handed off for delivery & payment
dispute resolved
Disputed
In Fulfillment
dispute raised
fulfillment & payment completed